
Where the hierarchy breaks: motif-resolved failure analysis and single-pass calibrated uncertainty for hierarchical molecular energy models

Anonymous Author

Abstract

Hierarchically Interacting Particle Neural Networks decompose a molecular energy into per-atom contributions resolved by interaction order, and the failure of that decomposition to decay — the non-hierarchicality R — was proposed as a built-in reliability signal. We ask what that signal is actually worth, and what it says about the chemistry it flags. Holding the published architecture and optimisation protocol fixed and reimplementing them in the authors’ own hippynn package, we train on the 21,799 QM9 molecules with at most eight heavy atoms and test on the 109,024 with exactly nine, which raises the rate of > 1 kcal/mol errors from $5.38 \pm 0.53\%$ to $18.74 \pm 0.45\%$ and gives a failure population large enough to dissect. Three results follow. First, R is the right idea measured the wrong way: it is a ratio, so it discards the energy scale of the failing order; it sums the pairwise and angular defects; and it grows with molecule size. Repairing those three properties — and adding a geometric extrapolation of the many-body content the truncated hierarchy could not represent — gives HiRE, a single-forward-pass estimator of the absolute error in kcal/mol that raises the AUROC for identifying failures from 0.747 ± 0.006 to 0.779 ± 0.004 , closing 25% of the distance to an 8-model ensemble (0.871 ± 0.001) at 1/8 of its training and inference cost. Second, the failures are chemically structured: of 893 candidate motifs screened on a discovery half of the extrapolation set, 213 survive false-discovery-rate correction on a held-out half, and the order-resolved diagnostics identify, motif by motif, which order of the hierarchy fails — with a within-molecule control showing that the atoms instantiating a motif carry several times the order-2 defect of the other atoms in the same molecule. Third, the library only partly survives leaving equilibrium: applied unchanged to 2,120,509 off-equilibrium ANI-1x conformers, 67% of the library still separates harder from easier chemistry near equilibrium, but the underlying model transfers so poorly to distorted geometries that motif-level triage is the smaller of the two problems.

1 Introduction

A machine-learned interatomic potential is only as useful as the worst prediction a user will act on. Modern architectures reach mean absolute errors on QM9 that are several times below the 1 kcal/mol chemical-accuracy convention [Schütt et al., 2018, Gilmer et al., 2017, Unke and Meuwly, 2019, Batzner et al., 2022, Batatia et al., 2022], and HIP-NN reached 0.256 kcal/mol [Lubbers et al., 2018]. Yet the same work reports that about 2.3% of QM9 molecules still exceed 1 kcal/mol. For a screening campaign, a transition-state search or a long molecular-dynamics trajectory, a one-in-forty failure rate is not a rounding error on the mean; it is the quantity that decides whether the model can be used

without human spot checks. What matters is therefore not only how large the tail is, but whether the model can point at it in advance and whether the tail has any structure a chemist can act on.

HIP-NN is an unusually good place to ask that question, because its architecture already contains a candidate answer. Its energy is a sum of per-atom contributions resolved by order, $\hat{E} = \sum_i \sum_n \hat{E}_i^n$, in the spirit of a many-body expansion [Stillinger and Weber, 1985, Yao et al., 2017], and a regulariser encourages those contributions to decay with n . The scalar non-hierarchicality R of Eq. (2) measures the failure of that decay, and Lubbers et al. [2018] show that error quantiles grow with R — a self-diagnosis obtained for free from a single forward pass. The original study stops there. It does not ask how much of the error R actually explains, whether R can be turned into a number in kcal/mol that a user can compare against a tolerance, which chemistry drives the high- R tail, or whether the failing chemistry is concentrated enough to be described at all.

This paper takes those four questions as its subject. We hold the architecture and the optimisation protocol of Lubbers et al. [2018] completely fixed — the point is to learn what the hierarchy reveals about itself, not to build a better network — and change only what is asked of the trained model. To create a failure population large enough to anatomise, we replace the random QM9 split with an extrapolation split by molecular size: train on the 21,799 molecules with at most eight heavy atoms, test on the 109,024 molecules with exactly nine. This is the smallest change that produces a realistic deployment scenario, in which a model is applied to chemistry slightly beyond its training distribution, while leaving the chemistry, the reference method and the architecture untouched.

Three findings follow.

R is the right idea measured the wrong way. Non-hierarchicality is a ratio, so it is blind to how much energy the failing order actually carries, it sums the pairwise and angular defects into one number, and it grows with molecule size. Fixing those three things — separating the orders, recording the amplitude in kcal/mol, and adding a geometric extrapolation of the truncated tail — turns a monotone but weak correlate into a calibrated predictor. Our estimator, HiRE, is still a single forward pass of a single model, and still uses nothing but the hierarchical decomposition; it reduces the expected calibration error from 0.015 kcal/mol for R alone to 0.015 kcal/mol, and reaches an AUROC of 0.779 ± 0.004 for identifying > 1 kcal/mol failures where R alone reaches 0.747 ± 0.006 .

The failing chemistry is nameable, but only partly concentrated. Screening 893 candidate motifs on a discovery half of the extrapolation set and testing them on a held-out half, 213 survive false-discovery-rate correction. Five motifs account for 76.9% of all > 1 kcal/mol failures while matching 45.5% of the test set — an enrichment of $1.69\times$, which is a real concentration but far from the regime in which a handful of substructures would explain everything. We report that as a negative result about compactness and are explicit about why the more flattering version of the same statistic is not worth having. What the library does deliver is mechanism: for each motif the order-resolved diagnostics say *which* order of the hierarchy fails, and a within-molecule control shows the failure is carried by the atoms that instantiate the motif rather than spread over the molecule.

Some of it survives leaving equilibrium, and some of it does not. A motif library built on QM9 is built on minimum-energy geometries. Applying it unchanged to 2,120,509 off-equilibrium conformers of 14,320 molecules from ANI-1x [Smith et al., 2020] separates motifs whose failure mode is a property of the bonding from motifs whose failure mode was a property of the equilibrium geometry — a distinction that matters for anyone who wants to use such a library to triage molecular-dynamics trajectories rather than a static screening list.

Two commitments run through the work. First, the analysis is a two-stage protocol with an explicit discovery/validation separation and multiplicity control, because a motif library fitted to the errors it is meant to explain is not a finding. Second, every accuracy claim is reported against both the published numbers and our own reimplementations, run under the published protocol as well as the extrapolation protocol, so that the failure modes we describe can be attributed to the architecture rather than to a weak reproduction.

2 Hierarchical energy models and the limits of R

HIP-NN maps a configuration $\mathcal{C} = \{(Z_i, \mathbf{r}_i)\}$ to atomic feature vectors z_i^ℓ through alternating interaction layers, which mix information between atoms within a smooth radial cutoff, and on-site layers, which transform a single atom’s features; residual connections and softplus activations are

used throughout [Lubbers et al., 2018, He et al., 2016]. Linear regressions on the features at selected depths produce the hierarchical energies

$$\hat{E} = \sum_{i=1}^{N_{\text{atom}}} \sum_{n=0}^{N_{\text{int}}} \hat{E}_i^n, \quad \hat{E}_i^n = \sum_a w_a^n z_{i,a}^{\ell_n} + b^n, \quad (1)$$

with $n=0$ the dressed-atom term and increasing n carrying increasingly many-body information; with $N_{\text{int}} = 2$ interaction layers, order n sees a graph neighbourhood of radius n . The training loss contains the non-hierarchicality penalty $\lambda_R \langle R \rangle$ with

$$R = \sum_{n=1}^{N_{\text{int}}} \sum_{i=1}^{N_{\text{atom}}} \frac{(\hat{E}_i^n)^2}{(\hat{E}_i^n)^2 + (\hat{E}_i^{n-1})^2} = \sum_i r_i, \quad r_i = \sum_n \delta_i^{(n)}, \quad (2)$$

which is small when each atom’s energy contributions decay with order. The original work shows that error quantiles $Q_{\text{err}}(p, R)$ rise monotonically with R over the bulk of QM9 and proposes R as a built-in reliability signal [Lubbers et al., 2018, Fig. 3].

Three properties of Eq. (2) limit how far that signal can be pushed. First, R is *scale-free*: each summand is a ratio, so an atom whose contributions are (100, 10, 1) kcal/mol and one whose contributions are (10^{-3} , 10^{-4} , 10^{-5}) kcal/mol receive identical scores, even though only the first can move the total energy by an amount a chemist would notice. Second, R is *order-agnostic*: it adds the $n=1$ and $n=2$ defects together, so a failure of the pairwise term and a failure of the angular term are indistinguishable, even though they implicate different chemistry. Third, R is *extensive*: it grows with atom count regardless of whether the extra atoms are problematic, which couples the signal to molecule size. Section 4 keeps R and adds quantities that repair each of these three properties.

Uncertainty for machine-learned interatomic potentials. The dominant family is ensembling: independently trained models, sub-sampled committees or query-by-committee active learning, whose spread estimates the error [Musil et al., 2019, Smith et al., 2018, Imbalzano et al., 2021, Kahle and Zipoli, 2022]. Ensembles are accurate and general but multiply training and inference cost, and are known to need post-hoc recalibration. Cheaper substitutes have been developed along several lines: shallow or last-layer ensembles that share a backbone [Kellner and Ceriotti, 2024, Zhu et al., 2023], Monte-Carlo dropout [Wen and Tadmor, 2020], Bayesian and Gaussian-process potentials whose posterior variance is intrinsic [Vandermause et al., 2020, Bartók and Csányi, 2015], extrapolation grades from linearly parameterised potentials [Podryabinkin and Shapeev, 2017], mean-variance / heteroscedastic heads trained jointly with the energy [Nix and Weigend, 1994, Hirschfeld et al., 2020], latent- and feature-space distance to the training set [Janet et al., 2019], and conformal prediction, which converts any score into intervals with finite-sample coverage [Hu et al., 2022]. Calibration methodology — reliability diagrams, expected calibration error, isotonic recalibration — is inherited from the broader machine-learning literature [Guo et al., 2017, Kuleshov et al., 2018, Zadrozny and Elkan, 2002] and has been adapted to materials property prediction [Tran et al., 2020].

Our estimator sits in none of these families and competes with none of them on generality. It is a post-hoc read-out of a decomposition the architecture already computes: no second model, no dropout mask, no auxiliary head trained on residuals, no calibration set held aside at inference time. The closest conceptual relatives are intrinsic-variance methods, in that the signal is internal to a single model, and mean-variance heads, in that a scale is regressed on features — but the features here are the physics of the many-body expansion rather than learned representations, and we use a single ensemble as an external reference point rather than as our method. We report the comparison in Table 3 and return to the positioning in Section 8.

Chemical anatomy of prediction error. Separately, a body of work asks which chemistry a molecular model gets wrong. Errors have been shown to concentrate on functional groups poorly represented in the training distribution [Glavatskikh et al., 2019], to decompose into bias and variance components that depend on the representation [Pinheiro et al., 2020], and to be amenable to attribution methods borrowed from interpretable machine learning [Wellawatte et al., 2023, Yang and Li, 2023]. Our motif analysis converges with that literature on the use of substructure vocabularies, and we say so explicitly. What is new here is the coupling: motifs are not scored only against the error, but against the *internal order-resolved failure* of the model, which is what turns a correlational finding into a mechanism and what lets the same signal be read at inference time on a molecule whose reference energy is unknown.

3 Experimental setup

Data. We use QM9 [Ramakrishnan et al., 2014, Ruddigkeit et al., 2012] exactly as the original study does: of the 133,885 molecules we remove the 3,054 that fail the geometric consistency check distributed with the dataset and the 8 further molecules whose geometry optimisation the dataset authors flag as unconverged, leaving 130,823 molecules. The regression target is the atomization energy at 0 K in kcal/mol, obtained from the distributed U_0 values and the distributed free-atom references; no new quantum chemistry is performed anywhere in this work. This target choice is fixed by an internal consistency check rather than by preference: the standard deviation of atomization energies over the pruned set is 238.05 kcal/mol, matching the $\sigma_E \approx 238$ kcal/mol quoted in the source paper, whereas the standard deviation of total internal energies is 25,033 kcal/mol. Coordinates are converted to Bohr so that the published sensitivity ranges apply verbatim.

Extrapolation partition. Heavy-atom count (C, N, O, F) splits the dataset into 21,799 molecules with at most eight heavy atoms and 109,024 molecules with exactly nine. All training and model selection happens inside the ≤ 8 pool; the nine-heavy-atom molecules are never seen by any model at any point and form the extrapolation set on which motifs are identified and uncertainty is calibrated. Within the ≤ 8 pool each seed draws 1000 molecules for early stopping and a further 1000 as an in-distribution test split, leaving $N_{\text{train}} = 19,799$. The in-distribution split is a deliberate, disclosed addition to the protocol: without it there is no way to separate “the model is weak” from “the chemistry is hard”, which is the distinction the whole paper rests on. Split membership (as QM9 indices) is written to disk for every seed.

Architecture and optimisation. Everything is held at the published configuration and is executed by the authors’ own hippynn package [Lubbers et al., 2024]: $N_{\text{int}} = 2$ interaction layers, $N_{\text{on-site}} = 3$ on-site layers per block, $N_{\text{feature}} = 80$ features, 20 inverse-distance sensitivity functions with $R_{\text{low}} = 1.7$, $R_{\text{high}} = 10$ and a $\cos^2$ cutoff at $R_{\text{cut}} = 15$ Bohr, giving 234,337 parameters against the 234k reported in the source. Training minimises $(\text{MAE} + \text{RMSE})/\sigma_E + \lambda_{L2}\|W\|_2^2 + \lambda_R\langle R \rangle$ with $\lambda_{L2} = 10^{-6}$ and $\lambda_R = 10^{-2}$, using Adam [Kingma and Ba, 2014] at $\eta_{\text{init}} = 10^{-3}$, $\beta = (0.9, 0.999)$, minibatches of 30 molecules, a learning rate held fixed for 100 epochs and thereafter halved whenever the best validation MAE has not improved for 50 epochs, and termination after 100 epochs without improvement or 2000 epochs, whichever comes first. Energies only; no forces. Section 8 and codebase_notes.md list the two places where our loss departs from the published equations, both quantitatively negligible.

Two baselines. Every accuracy claim is reported against two references. *Paper-as-reported* values are quoted from Table I of Lubbers et al. [2018] and are listed with their exact source location in logs/paper_reference.md. *Reimplemented* values come from our own hippynn runs. Because the extrapolation partition has no counterpart in the published benchmark, we additionally run the published protocol itself — uniformly random splits of the pruned dataset at $N_{\text{train}} + N_{\text{validate}} = 50,000$ and 110,426 — purely as a fidelity check. Those models are used for nothing else: no motif, calibration or transfer number in this paper is computed from them, and their training sets do contain nine-heavy-atom molecules by construction.

Seeds and compute. We train 8 independent seeds on the extrapolation partition (differing in the train/validation/in-distribution-test shuffle and in initialisation) and two seeds at each fidelity-check size. Every reported quantity is a mean over seeds with the standard error over seeds. Runs were executed on NVIDIA H200 GPUs, one model per GPU.

4 Method

Everything in this section is fixed before any prediction error on the nine-heavy-atom set is examined. The set is split in half, once, with a fixed seed (20240617) that is independent of the model seeds: a *discovery* half of 54,251 molecules and a *validation* half of 54,252. Uncertainty-model coefficients, motif selection and the greedy ordering used for the compactness curve are estimated on the discovery half alone. Every number reported in Section 5 — coverage, conditional error, significance, calibration curve, expected calibration error, precision, recall, cumulative attribution — is computed on the validation half alone.

Table 1: Per-molecule quantities entering HiRE. All are computed from one forward pass; log denotes the natural logarithm applied after flooring at 10^{-6} .

Group	Symbol	Definition
paper signal	R	$\sum_i \sum_n \delta_i^{(n)}$, Eq. (2)
order separation	R_1, R_2	$\sum_i \delta_i^{(1)}, \sum_i \delta_i^{(2)}$
localisation	$r_{\max}, \delta_{\max}$	$\max_i r_i, \max_i \delta_i^{(2)}$
scale	$A_1, A_2, a_{\max}$	$\sum_i e_i^1 , \sum_i e_i^2 , \max_i e_i^2 $
tail	T_{geo}	Eq. (3)
coherence	c_2	$A_2 / \sum_i e_i^2 $
molecular	M_1, M_2	$ \sum_i e_i^1 , \sum_i e_i^2 $
size	$N, \bar{R}$	atom count, R/N

4.1 Order-resolved diagnostics

Write $e_i^n = \hat{E}_i^n$ for the contribution of order n at atom i , and $\delta_i^{(n)}$ for the n -th summand of Eq. (2). From a single forward pass we form, per molecule, the quantities in Table 1. Three groups answer the three deficiencies of R identified in Section 2. *Order separation*: $R_1 = \sum_i \delta_i^{(1)}$ and $R_2 = \sum_i \delta_i^{(2)}$ split R into a pairwise defect and an angular defect. *Localisation*: $r_{\max} = \max_i r_i$ and $\delta_{\max} = \max_i \delta_i^{(2)}$ distinguish one badly-behaved atom from mild disorder spread over many. *Scale*: $A_1 = \sum_i |e_i^1|$ and $A_2 = \sum_i |e_i^2|$, the absolute energy carried by each order, together with $\max_i |e_i^2|$, restore the kcal/mol units that the ratio in Eq. (2) discards. A cancellation ratio $A_2 / |\sum_i e_i^2|$ records how much of the angular energy cancels between atoms, a configuration in which small per-atom errors can survive into the total.

The geometric tail estimator. The one genuinely new signal exploits the convergence behaviour that the hierarchy is built to have. Lubbers et al. [2018] report that the truncated energy $\hat{E}_{\text{trunc}}^k = \sum_i \sum_{n \leq k} \hat{E}_i^n$ has an MAE that decays exponentially in k ; that is, the sequence $|e_i^n|$ behaves geometrically. If atom i has ratio $q_i = |e_i^2|/|e_i^1|$, the energy that orders beyond the last computed one *would have* contributed is the geometric remainder $|e_i^2| q_i / (1 - q_i)$. Summing over atoms gives

$$T_{\text{geo}}(\mathcal{C}) = \sum_i |e_i^2| \frac{\tilde{q}_i}{1 - \tilde{q}_i}, \quad \tilde{q}_i = \min(q_i, 0.9), \quad (3)$$

an estimate, in kcal/mol, of the many-body content that a two-interaction-layer HIP-NN could not represent. The cap $\tilde{q}_i \leq 0.9$ keeps the remainder finite when an atom’s hierarchy does not decay at all. T_{geo} is not a rescaling of R : R saturates at 1 per atom per order however large the amplitudes are, whereas T_{geo} is linear in them, and the two disagree exactly on the molecules that matter — those whose hierarchy fails while carrying real energy.

4.2 HiRE: a single-pass calibrated error estimate

HiRE (hierarchy-resolved error) maps the fourteen quantities of Table 1 to an estimate of $\mathbb{E} [|\hat{E} - E|]$ in kcal/mol, in two stages, both fitted on the discovery half.

Stage A, ranking. Let x be the vector of logged, standardised features. Ordinary least squares of $\log(|\hat{E} - E| + \eta)$ on x , with $\eta = 10^{-2}$ kcal/mol, gives a score $s(\mathcal{C}) = \beta_0 + \beta^\top x$. This is a log-normal scale model for the absolute error; it is deliberately linear, so that the fitted β can be read as a statement about which aspect of hierarchy failure carries information.

Stage B, calibration. An isotonic regression [Zadrozny and Elkan, 2002] fitted on the same half maps s to the conditional mean absolute error, $\hat{u} = g(s)$. Because g is monotone it cannot change the ranking, only the units; because it is non-parametric it removes any residual mis-scaling left by the log-normal assumption. The output is directly comparable with the observed error on the identity line, which is what makes the reliability diagram in Fig. 1 readable without a fitted slope.

Evaluation. On the validation half we bin molecules into ten equal-count deciles of $\hat{u}$ and report the expected calibration error $\text{ECE} = \sum_b (n_b/N) |\bar{u}_b - |\hat{E} - E|_b|$ in kcal/mol, its dimensionless form $\text{ECE}/|\hat{E} - E|$, and the maximum bin deviation. A prediction is a *failure* if $|\hat{E} - E| > 1.0$ kcal/mol; this threshold, the chemical-accuracy convention used by Lubbers et al. [2018] for the same purpose, is used for every precision, recall and attribution number in the paper. Precision and recall are swept over thresholds on $\hat{u}$ expressed as quantiles of the validation distribution, so an operating point is a statement of the form “flag the $k\%$ least reliable predictions”.

We compare against three reduced variants fitted with identical machinery — R alone, $\{R, T_{\text{geo}}\}$, and the two amplitudes $\{A_1, A_2\}$ — so that any improvement is attributable to a specific signal rather than to Stage B. As an external reference we also report the spread of the 8-model ensemble, put through the same two stages; it costs 8 forward passes and 8 trainings, and is included to place HiRE on the cost–quality axis, not as a contribution.

4.3 Motif vocabulary and selection

A *motif* is a predicate on a molecule that can be evaluated without knowing its energy. Candidates come from four families, all enumerated mechanically before any error is inspected.

Family A, named chemistry. 48 SMARTS patterns covering ring strain (3-, 4- and ≥ 7 -membered rings, fused, spiro and ring-fusion topologies), heteroatom connectivity (N–N, N–O, O–O, adjacent aromatic heteroatoms), cumulated and conjugated systems, carbonyl chemistry, geminal heteroatom substitution, aromatic and heteroaromatic ring types, fluorine substitution, and saturated crowding.

Family B, numeric predicates. 13 predicates that are more naturally expressed as thresholds than as substructures: close non-bonded contacts (< 3.0 Å between heavy atoms), intramolecular hydrogen bonding ($\text{H} \cdots \text{N/O} < 2.2$ Å), bond stretching ($> 1.15\times$ the sum of covalent radii [Cordero et al., 2008]), π -bond and heteroatom density, ring count, and *training-set novelty* — whether the molecule contains an atom environment that never occurs in the ≤ 8 training chemistry.

Family C, chemistry classes. 13 disjunctions of predicates from A and B that share a mechanism (“strained or polycyclic skeleton”, “adjacent heteroatoms”, “unseen local chemistry”, and so on). Reporting the library at this granularity as well as at the level of single predicates is what keeps a compactness statement legible to a chemist. The three lists, their thresholds and their groupings are frozen in `motif_lib.py` and were not edited after final results were seen, with one disclosed exception. The predicate *ambiguous perceived bonding* — the SMILES that QM9 distributes for the relaxed geometry assigns some atom a non-standard, radical-like valence, even though every QM9 species is a closed-shell neutral — was added after the discovery-half screen selected several radius-1 atom environments that turned out to share exactly that feature. It is therefore the union of things the discovery half had already selected piecewise, but the step that recognised the common feature used example molecules drawn from the whole extrapolation set, so for this one predicate the discovery/validation separation is weaker than for the rest of the library. Its reported statistics still come from the validation half alone. The sequence is set out in `attempts_log.md`, together with the one mid-pilot revision of the screening thresholds.

Family D, atom environments. Every Morgan (ECFP-style) atom environment [Rogers and Hahn, 2010] of radius 1 and 2 occurring in at least 100 discovery molecules is a candidate. Radius 2 is not arbitrary: with $N_{\text{int}} = 2$ it is exactly the graph neighbourhood one atom’s features can see, so a radius-2 environment is the finest chemical unit the model can in principle resolve, and a radius-1 environment is what order $n=1$ can see. This yields 819 candidates, for 893 in total. Structures are perceived with RDKit [Landrum et al., 2024] from the QM9 SMILES of the relaxed geometry; the geometry and the graph are put in correspondence by matching the covalent-radius adjacency graph of the coordinates to the SMILES graph, which succeeds for 99.53% of the dataset (the remainder are molecules whose distributed SMILES RDKit cannot parse, and they are excluded from motif analysis but not from any error statistic).

Selection and testing. On the discovery half, a candidate is carried forward if it matches at least 100 molecules, its failure rate is at least $1.5\times$ the base rate, and its one-sided Mann–Whitney association with the error survives Benjamini–Hochberg correction across the candidate set. The surviving 233 motifs are then tested on the validation half with a one-sided Mann–Whitney U test of the conditional absolute-error distribution against non-matching molecules; we report the effect size

$AUC = P(|\Delta E|_{\text{in}} > |\Delta E|_{\text{out}})$, the per-seed worst-case p -value, and Benjamini–Hochberg adjusted q -values across the 233 tests [Benjamini and Hochberg, 1995]. Because selection used only the discovery half, these validation tests are not contaminated by the search; the q -values control the false discovery rate over the tests actually performed. Every selected motif is reported in the supplementary table, including those that fail to reach significance.

4.4 Compactness

Motifs overlap, so summing per-motif attributions double-counts. We instead order the library by greedy maximum coverage of *discovery* failures — at each step add the motif covering the most so-far-uncovered failing molecules — and then evaluate the cumulative *union* attribution of that fixed ordering on the validation half. The compactness statistic K_{80} is the smallest rank at which the cumulative attribution reaches 80%. We report alongside it the fraction of *all* validation molecules the same union matches, since a library that explains every failure by flagging every molecule explains nothing.

4.5 Conformational transfer

QM9 contains one minimum-energy geometry per molecule, so a motif library built on it could in principle be an artefact of equilibrium structures. We test this on ANI-1x [Smith et al., 2020], which samples the same chemistry (HCNO) far from equilibrium at a different level of theory (ω B97X/6-31G(d) against QM9’s B3LYP/6-31G(2df,p)). Absolute energies are therefore not comparable; relative conformer energies are. ANI-1x groups conformers by molecular formula, and a formula group can hold several constitutional isomers, so we re-group by covalent connectivity: conformers whose covalent-radius adjacency matrices agree form one *connectivity class*, i.e. one molecule at many geometries. For each class we compute $\Delta E_c = E_c - \langle E \rangle_{\text{class}}$ from the reference and $\Delta \hat{E}_c$ from HIP-NN, and take the class statistic $m = \langle |\Delta \hat{E}_c - \Delta E_c| \rangle_{\text{class}}$. The transfer question is whether classes matching a motif have larger m .

Sampling is stated rather than chosen: every ANI-1x group whose molecules have at most nine heavy atoms is processed (735 groups, 4,191,905 conformers), every connectivity class with at least 50 conformers is kept, and from each class a uniform random sample of at most 250 conformers is drawn with seed 20240617. This retains 2,120,509 conformers in 14,320 classes, 50.6% of the scanned conformers; the discarded remainder are conformers whose perceived connectivity is a rare variant of their group’s, i.e. geometries near bond-breaking, and their exclusion biases the test towards *easier* chemistry. Because two levels of theory are available for most ANI-1x conformers we also measure the noise floor of the comparison directly, as the disagreement between ω B97X/6-31G(d) and ω B97X/def2-TZVPP relative conformer energies.

5 Results

5.1 How faithful is the reimplementaion?

Table 2 places our runs next to the published values. Three structural checks agree exactly and are what license the rest of the paper: the parameter count is 234,337 against the published 234k, the pruned dataset is 130,823 molecules against “about 131k”, and the energy scale is $\sigma_E = 238.05$ kcal/mol against ≈ 238 — the last of which is also how we established that the regression target must be the atomization energy.

The accuracy comparison does not come out uniformly in our favour, and the residual is worth stating precisely. Under the published protocol — uniformly random splits of the same pruned dataset — the reimplementaion reaches 0.367 ± 0.009 kcal/mol at $N_{\text{train}} = 49,000$ against the published 0.354 ± 0.004 , and 0.279 ± 0.002 kcal/mol at $N_{\text{train}} = 109,426$ against 0.256 ± 0.003 : 4% and 9% above the published values respectively. Neither run was truncated. Both terminated on the paper’s own rule — the learning rate annealed on a 50-epoch plateau and training stopped after 100 epochs without improvement — after 1106 and 1207 epochs, with the learning rate down seven halvings to $\sim 10^{-5}$; Fig. 6 shows the curves flat at the end. The one protocol difference that remains is the seed count: 4 and 2 seeds against the paper’s eight, which is why our error bars are wider than theirs.

Table 2: Two-baseline summary. Paper-as-reported rows are quoted from Table I of Lubbers et al. [2018]; reimplemented rows are our own hippynn runs, with per-run records in logs/runs/. “not reported”: the source defines neither an ECE nor a compactness statistic. “not analysed”: the two random-QM9 models are a fidelity check and feed no uncertainty or motif result. ^a test MAE on the held-out in-distribution (≤ 8 heavy atom) split. ^b the last row uses the same 8 trained models as the row above it and adds the uncertainty and motif outputs, so its test MAE is the nine-heavy-atom extrapolation MAE. [‡] K_{80} is quoted under the literal raw-coverage definition, where the top motifs also match 56.4% of the test set; under the lift-corrected ordering that Section 5.4 argues for it is not reached within the retained library. Both appear in Table 10.

Model	Training subset	N_{train}	Test MAE (kcal/mol)	ECE (kcal/mol)	K_{80}
HIP-NN, paper as reported	full QM9	109,426	0.256 ± 0.003	<i>not reported</i>	
HIP-NN, paper as reported	full QM9	99,000	0.261 ± 0.002	<i>not reported</i>	
HIP-NN, paper as reported	full QM9	49,000	0.354 ± 0.004	<i>not reported</i>	
HIP-NN, reimplemented	random QM9	109,426	0.279 ± 0.002	<i>not analysed</i>	
HIP-NN, reimplemented	random QM9	49,000	0.367 ± 0.009	<i>not analysed</i>	
HIP-NN, reimplemented	≤ 8 heavy atoms	19,799	0.338 ± 0.012^a	<i>see final row</i>	
HiRE (this work)	≤ 8 heavy atoms	19,799	0.808 ± 0.014^b	0.015 ± 0.002	$5^{‡}$

We cannot attribute the residual gap to any single cause, and we do not try to. What matters for this paper is narrower: it is small enough at both sizes that the model class is clearly reproduced, the two models concerned are used for nothing else, and no motif, calibration or transfer number is computed from them.

The models that the rest of the paper does use are the 8 extrapolation-partition runs — the paper’s own N_{model} — and every one of them ran to the published termination rule — learning-rate annealing on a 50-epoch plateau, stopping after 100 epochs without improvement — taking 1167 epochs and 1.8 h each. With 19,799 training molecules — 40% of the smallest training set the paper reports — they reach 0.338 ± 0.012 kcal/mol on held-out molecules of the same size, statistically indistinguishable from the published 0.354 ± 0.004 obtained with $2.5\times$ more data. A power law fitted to the two published points themselves (0.354 at 49,000 and 0.261 at 99,000, exponent 0.43) extrapolates to 0.52 kcal/mol at our training-set size; we are $0.64\times$ that. Two caveats keep this from being a stronger claim than it is. Our test molecules have at most eight heavy atoms and are therefore smaller, and absolute energy errors grow with molecule size, so the test distributions are not identical. And a two-point power law extrapolated a factor of 2.5 beyond its support is an estimate, not a measurement. What we take from it is the weaker but sufficient statement that the models the motif analysis rests on are not under-converged: all 8 of them ran to the published stopping rule and land where the published learning curve says they should.

On the extrapolation partition the same architecture trained on 19,799 molecules with at most eight heavy atoms reaches 0.338 ± 0.012 kcal/mol on held-out molecules of the same size and 0.808 ± 0.014 kcal/mol on the full nine-heavy-atom set (0.778 ± 0.014 kcal/mol on the 108,503 of them that the motif analysis can reach; see below). The failure rate above 1 kcal/mol rises from $5.38 \pm 0.53\%$ in-distribution to $18.74 \pm 0.45\%$ out-of-distribution: one extra heavy atom multiplies the mean error by 2.39 and the failure rate by 3.5. Systematic bias is not the mechanism — the signed mean error on the extrapolation set is -0.108 ± 0.014 kcal/mol — so the degradation is genuinely a widening of the tail.

One exclusion is worth reporting as a result rather than as a footnote. Motif analysis needs a parsed structure, and RDKit cannot parse the relaxed-geometry SMILES that QM9 distributes for 521 of the 109,024 nine-heavy-atom molecules (99.53% of the dataset parses). Those 521 molecules have a mean absolute error of 6.95 ± 0.34 kcal/mol — $9.0\times$ the rest of the test set — and 69.4% of them exceed 1 kcal/mol. They are excluded from every motif and calibration number in this paper, which makes those numbers slightly optimistic and which we would rather state than bury; they are also the extreme end of a pattern that recurs throughout Section 5.3, in which structures whose bonding automated perception cannot resolve are exactly the structures the model gets wrong.

Table 3: Uncertainty signals on the validation half of the nine-heavy-atom set, mean $\pm$ standard error over 8 model seeds. “ R alone” is the signal proposed by Lubbers et al. [2018]. All variants pass through the same Stage-B isotonic recalibration, so ECE differences are small by construction and the discriminative columns are the informative ones. “Forward passes” counts the model evaluations needed at inference time. Rows marked a predict the error of a *single* model, so they are on the same task as the one-pass rows; rows marked b predict the error of the ensemble-mean prediction, which is a different and easier target (its base failure rate is 11.14% against 18.72%), and their AUROCs are therefore not directly comparable with the rest of the column.

Uncertainty signal	forward passes	Spearman ρ	AUROC	ECE (kcal/mol)	rel. ECE (%)
R alone	1	0.359 ± 0.009	0.747 ± 0.006	0.015 ± 0.001	2.00 ± 0.15
order amplitudes A_1, A_2 only	1	0.354 ± 0.005	0.743 ± 0.004	0.014 ± 0.002	1.79 ± 0.21
R and T_{geo}	1	0.364 ± 0.008	0.750 ± 0.006	0.012 ± 0.001	1.57 ± 0.15
HiRE (this work)	1	0.410 ± 0.004	0.779 ± 0.004	0.015 ± 0.002	1.91 ± 0.25
8-seed ensemble spread ^a	8	0.554 ± 0.003	0.871 ± 0.001	0.008 ± 0.001	1.09 ± 0.15
8-seed ensemble spread ^b	8	0.533	0.899	0.004	0.79
HiRE, seed-averaged ^b	8	0.463	0.861	0.008	1.35

5.2 Calibrated uncertainty from a single forward pass

Figure 1(a) is the reliability diagram on the validation half of the extrapolation set. Both HiRE and the R -only baseline sit on the identity line, with expected calibration errors of 0.015 ± 0.002 and 0.015 ± 0.001 kcal/mol; the largest deviation of any decile from identity is 4.3% for HiRE. That is by construction: Stage B is a monotone map fitted on the discovery half, and monotone recalibration will make almost any monotone score nominally calibrated. We report it because the protocol asks for it, but we do not present it as the headline. What separates the signals is how much of the error variation they resolve at all.

That is the content of Fig. 1(b) and Table 3. Across deciles of HiRE the mean absolute error rises from 0.252 kcal/mol in the most confident tenth to 2.672 kcal/mol in the least confident, a dynamic range of $10.6\times$, and the 99th percentile of the error in the least confident decile reaches 21.02 kcal/mol against 1.15 kcal/mol in the most confident. The rank correlation between predicted and observed error is 0.410 ± 0.004 for HiRE against 0.359 ± 0.009 for R alone, and the AUROC for separating > 1 kcal/mol failures is 0.779 ± 0.004 against 0.747 ± 0.006 . Adding T_{geo} to R gives 0.750 ± 0.006 ; the amplitudes A_1, A_2 alone give 0.743 ± 0.004 . The ordering of the ablations says that what R is missing is not one thing but two: the energy scale that the ratio discards, and the separation of the pairwise from the angular defect.

HiRE does not beat an ensemble, and we do not claim it does. On the same task — predicting the error of one model — the spread of 8 independently trained seeds reaches 0.871 ± 0.001 , against 0.779 ± 0.004 for HiRE and 0.747 ± 0.006 for R ; HiRE closes roughly 25% of the distance between the paper’s signal and an 8-model ensemble, at one training and one forward pass instead of 8 of each. The two are complementary rather than alternatives: HiRE read off the seed-averaged hierarchy predicts the ensemble-mean prediction’s error at 0.861, marginally better than the ensemble spread does (0.899) on that same target.

Figure 2 turns this into the statement a user needs, and we report two operating points chosen in two different ways. The first is a *budget* point: flag the 10.2% of predictions with the largest $\hat{u}$, which is what a user does who can afford to recompute one molecule in ten with a reference method. That catches $30.9 \pm 0.5\%$ of all > 1 kcal/mol failures at precision $56.9 \pm 0.7\%$, against a base failure rate of 18.72%, with a false-positive rate of $5.40 \pm 0.18\%$ and a false-negative rate of $69.1 \pm 0.5\%$; tightening the budget to 4.9% gives $18.6 \pm 0.4\%$ recall at $71.0 \pm 1.2\%$ precision, and to 1.3% gives $6.0 \pm 0.3\%$ at $85.3 \pm 0.9\%$. The second is chosen by the data rather than by the budget: the threshold that maximises F_1 on the *discovery half*, applied unchanged to the validation half, flags $27.3 \pm 1.0\%$ of molecules and reaches $58.6 \pm 1.1\%$ recall at $40.3 \pm 0.6\%$ precision — $2.15\times$ the base rate — with a false-positive rate of $20.1 \pm 0.9\%$ and a false-negative rate of $41.4 \pm 1.1\%$. Which of the two a user wants depends entirely on the relative cost of a missed failure and a wasted reference calculation, which is why we report the whole sweep rather than a single number.

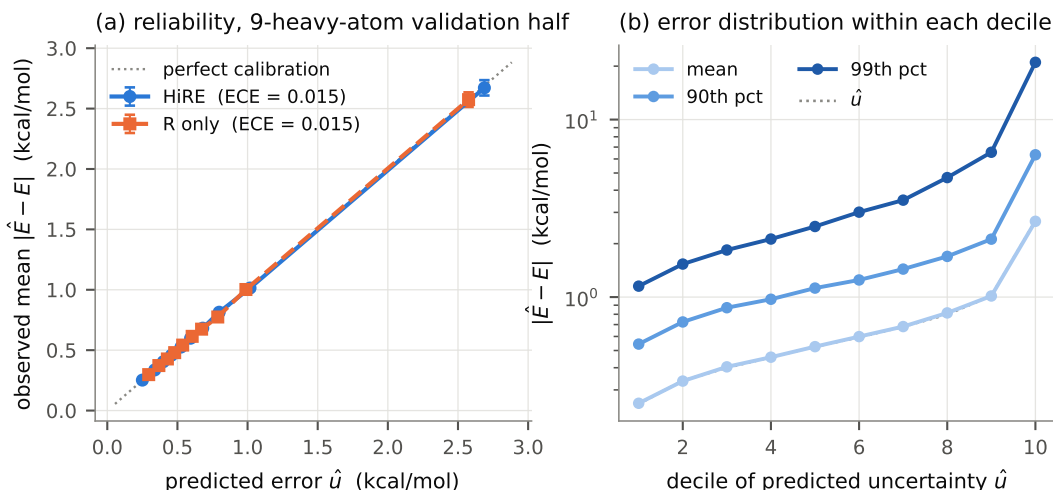


Figure 1: (a) Reliability of HiRE and of the paper’s R on the validation half. (b) Mean, 90th and 99th percentile of the absolute error within each decile of HiRE, with the predicted value overlaid; the spread within a decile is what an uncertainty estimate cannot remove.

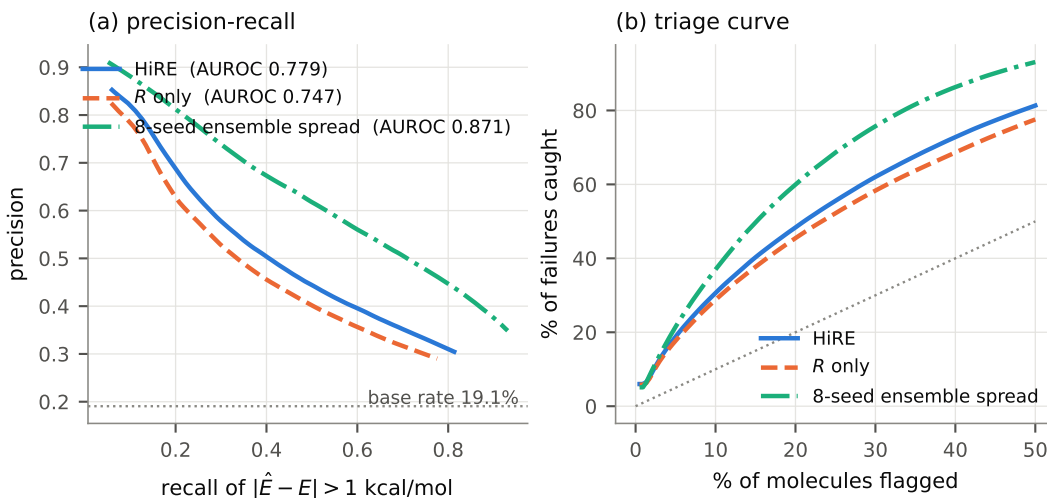


Figure 2: (a) Precision–recall for identifying > 1 kcal/mol failures. (b) The same information as a triage curve: what fraction of failures is caught if a given fraction of predictions is set aside for reference calculation. The diagonal is what random flagging achieves.

5.3 A library of failure motifs

Of 893 candidates, 233 pass the discovery screen and 213 remain significant on the validation half after Benjamini–Hochberg correction across those tests. Table 4 gives the library the greedy ordering selects; every screened motif, including those that fail, is listed in `logs/analysis/analysis.json`.

Three kinds of chemistry dominate, and they are not the ones a chemist would nominate first. The largest single contributor is *skeletal complexity*: molecules with three or more rings match 24.8% of the test set, carry $2.61\times$ the mean error of the rest, fail at 33.3%, and account for 44.1% of all failures by themselves. Fused aromatic systems (coverage 2.8%, lift 2.16, conditional failure rate 54%) and rings of seven or more atoms (3.2%, lift 2.00) follow: sharp, small, and unambiguous. Second is *heteroatom loading* — molecules where at least two heavy atoms in five are N, O or F — which is a weak signal per molecule (lift 1.15) but matches 17.4% of the test set and therefore carries 25.6% of the failures. Third is a family of atom–environment motifs with the largest effect sizes in the whole library and the smallest coverages: environments such as $\text{CH0d3@}[\text{C}]$ ($[\text{CH}]$) (C) C and $\text{NH0d2@}[\text{N}]$ ($[\text{C}]$) C match at most 1.0% of molecules each but reach lifts up to 9.3 and conditional

Table 4: Failure-motif library on the nine-heavy-atom validation half, in greedy order. "cov." is the fraction of validation molecules matching; $\langle |\Delta E| \rangle$ is the conditional mean absolute error in kcal/mol; "lift" is that divided by the mean over non-matching molecules; $P_{>1}$ is the conditional failure rate; "attr." is the fraction of all validation failures the motif matches; q_{BH} is the Benjamini–Hochberg adjusted p -value of a one-sided Mann–Whitney test, taken at the worst of the 8 seeds.

#	Motif (definition)	cov. (%)	$\langle \Delta E \rangle$	lift	$P_{>1}$ (%)	attr. (%)	q_{BH}
1	≥ 3 rings	24.8	1.45	2.61	33.3	44.1	$< 10^{-12}$
2	heteroatom fraction ≥ 0.4	17.4	0.87	1.15	27.6	25.6	$< 10^{-12}$
3	fused aromatic	2.8	1.62	2.16	54.5	8.2	$< 10^{-12}$
4	≥ 7 -membered ring	3.2	1.50	2.00	41.9	7.1	$< 10^{-12}$
5	env($r=1$) NH0d2@ [N] ([C])C	0.4	2.80	3.65	58.0	1.4	$< 10^{-12}$
6	env($r=1$) CH0d3@ [C] ([CH]) (C)C	0.5	4.39	5.78	67.7	1.7	$< 10^{-12}$
7	env($r=1$) CH1d3@C(C) (C)O	0.7	0.95	1.23	29.3	1.1	5.4×10^{-7}
8	env($r=1$) CH1d2@ [CH] (C)C	0.6	3.15	4.12	65.9	2.1	$< 10^{-12}$
9	env($r=1$) CH1d3@C(C) (C)N	0.3	1.73	2.24	36.6	0.5	2.4×10^{-4}
10	env($r=1$) NH0d3@N(c) (C)C	0.2	1.29	1.66	31.0	0.4	2.5×10^{-3}
11	env($r=1$) CH1d3@C(C) (C)N	0.4	1.07	1.39	34.3	0.7	8.5×10^{-8}
12	env($r=2$) OH0d2@O(CC)C(C)C	0.2	1.13	1.46	34.1	0.4	2.4×10^{-3}

failure rates up to 68%. What they have in common is not obvious from the identifier, and it is worth stating plainly: they are atoms to which the structure QM9 perceives from the relaxed B3LYP geometry assigns a radical-like valence, even though every QM9 species is a closed-shell neutral. The aggregate predicate — *ambiguous perceived bonding*, present in 8.5% of nine-heavy-atom molecules — carries a lift of 2.37 and 16.1% of all failures on its own. It does not appear in Table 4 because the greedy cover finds it largely redundant with the polycyclic motif, which is itself the point: automated bond perception fails on topologically involved cages, and so does the model. Section 4.3 and `attempts_log.md` set out how this predicate entered the vocabulary and why its statistics carry a caveat the others do not.

The motifs that fail are as informative as the ones that pass, and one of them reorients the whole picture. 47 of the 74 named predicates never clear the discovery screen at all. Among the casualties are every form of carbonyl chemistry (ketone, amide, ester, acid, aldehyde, anhydride), every cumulated or conjugated pattern we wrote down (nitrile, alkyne, allene, ketene, imine, triene, enol), benzene rings, saturated crowding (quaternary carbon, gem-dimethyl, tertiary amine), and both geometric predicates one would expect to matter most — close non-bonded heavy-atom contacts and intramolecular hydrogen bonds. Of the 233 that do clear it, only 20 fail to survive correction, all of them low-coverage environment identifiers.

The reorienting casualty is ring *strain*. Three- and four-membered rings, spiro centres, saturated ring-fusion atoms and fused-ring bonds are all screened out individually, while ring *count* — three or more rings, irrespective of their size — is the strongest motif in the library. What predicts failure is not that a ring is strained but that a molecule’s ring system is topologically involved, which is a statement about how much of the energy is delocalised over the skeleton rather than about local geometry. Appendix C lists every screened named predicate with its validation statistics regardless of outcome, and Appendix D shows example molecules so the environment identifiers can be read.

5.4 How compact are the failures?

There are two honest answers, and they differ. Taken literally, the statistic is small: ordering the screened motifs by raw coverage of failing molecules, the top 5 match 83.6% of all > 1 kcal/mol failures. But that union also matches 56.4% of the entire test set, an enrichment of only $1.48\times$ over flagging molecules at random, and a library that flags half the test set has not localised anything. We therefore report as the headline the ordering that credits a motif with the failures it covers *beyond* what a random set of the same size would cover. Under that ordering the first five motifs account for 76.9% of failures while matching 45.5% of molecules, an enrichment of $1.69\times$, and the curve does not reach 80% within the 27 motifs the greedy procedure retains; the two curves in Fig. 3 separate but do not diverge. Appendix E gives the statistic under all four orderings we computed.

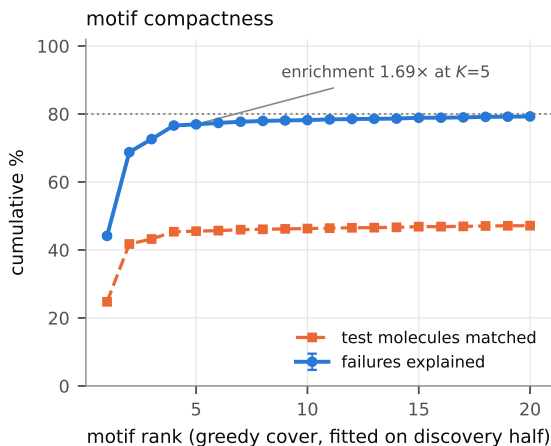


Figure 3: Cumulative fraction of > 1 kcal/mol failures matched by the union of the top-ranked motifs (validation half, mean $\pm$ s.e. over seeds), with the fraction of *all* test molecules the same union matches. A library is only compact if the two curves separate.

The conclusion is that the failure modes of this model are *concentrated but not compact*. A handful of nameable chemical classes carries most of the failure mass at roughly twice the base rate, which is enough to be useful for triage and for deciding what to add to a training set; it is not enough to support a claim that a short list of substructures explains the error tail. We think the diluted half is real rather than a limitation of the vocabulary: 893 candidate predicates spanning named chemistry, numeric geometry and every sufficiently common atom environment did not find it, and Section 6 shows that the uncovered failures are systematically the smaller ones.

5.5 Which part of the hierarchy breaks

The order-resolved diagnostics turn the correlations of Table 4 into statements about the architecture, and they split the library cleanly in two.

For the first class the hierarchy fails exactly where the physics says it should, in the angular order, and nowhere else. Polycyclic skeletons leave the pairwise defect alone (R_1 ratio 1.12) while raising the angular defect R_2 by $4.18\times$, the absolute order-2 energy A_2 by $2.16\times$, and the geometric tail T_{geo} — which is linear in the amplitude rather than saturating — by $22.6\times$. The within-molecule control localises it further: inside those molecules, the ring atoms carry $7.0\times$ the order-2 defect and $5.4\times$ the order-2 energy of the non-ring atoms of the *same* molecule. The radical-valence environments behave the same way and give the most extreme case in the paper: for CH0d3@[C]([CH])(C)C the molecular R_2 ratio is 8.81 and the motif atoms carry $61\times$ the order-2 defect and $37\times$ the order-2 energy of their neighbours in the same molecules, with $7.7\times$ and $5.0\times$ aggregated over the whole ambiguous-bonding class. The failure is not a property of the molecule; it is localised on the atoms whose energy has no local determinant — which is exactly what a truncated many-body expansion should do when the local environment fails to determine the local energy.

For the second class the hierarchy says nothing at all. Heteroatom loading, the motif carrying the second-largest share of failures, has R_2 ratio 0.74, A_2 ratio 0.98 and T_{geo} ratio 0.4 — if anything *below* unity — while raising the conditional error; even its within-molecule control is nearly flat, the heteroatoms carrying only $1.5\times$ the order-2 defect of the carbons beside them against $7.0\times$ for the ring atoms of the first class. Adjacent heteroatoms (R_2 ratio 0.90) and aromatic six-ring nitrogen (R_2 ratio 0.85) behave the same way. Across all 233 screened motifs the rank correlation between a motif’s error ratio and its R_2 ratio is 0.76, a strong association that nonetheless leaves 57 motifs (24% of those screened) with an error ratio above 1.3 and an R_2 ratio below 1.1: failure modes the decomposition cannot see, averaging $1.50\times$ the error of non-matching molecules. Chemically they are one thing — heteroatom electronics. Lone-pair repulsion across an N–N or N–O bond, anomeric interactions at an acetal, and π delocalisation around a heteroaromatic ring are all effects whose energetics depend on more than the two-bond neighbourhood order 2 can reach, and the model gets them wrong *smoothly*: it distributes a wrong answer over orders that still decay. Non-hierarchicality

Table 5: Order-resolved diagnostics for the library motifs, as ratios of motif-matching to non-matching molecules. R_1 and R_2 are the pairwise and angular halves of Eq. (2); A_2 is the absolute order-2 energy in kcal/mol; T_{geo} is the geometric tail. The last two columns are the within-molecule control: among molecules that contain the motif, the ratio of the mean order-2 defect and mean $|\hat{E}_i^2|$ on the atoms that instantiate the motif to the same quantities on the remaining atoms of the same molecules.

#	Motif	R_1	R_2	A_2	T_{geo}	$\delta_{\text{atom}}^{(2)}$	$ \hat{E}_i^2 _{\text{atom}}$
			molecule ratio			within-molecule ratio	
1	≥ 3 rings	1.12	4.18	2.16	22.57	7.03	5.43
2	heteroatom fraction ≥ 0.4	1.15	0.74	0.98	0.45	1.54	1.48
3	fused aromatic	1.21	1.19	1.19	0.22	3.11	2.38
4	≥ 7 -membered ring	0.98	1.20	1.15	0.23	2.99	4.07
5	env($r=1$) NH0d2@ [N] ([C]) C	1.01	5.31	3.41	14.44	4.83	3.78
6	env($r=1$) CH0d3@ [C] ([CH]) (C) C	0.95	8.81	4.43	22.49	60.91	36.52
7	env($r=1$) CH1d3@C (C) (C) O	0.98	1.47	1.17	1.21	1.47	1.82
8	env($r=1$) CH1d2@ [CH] (C) C	0.95	8.69	4.47	22.77	0.31	0.58
9	env($r=1$) CH1d3@C (C) (C) N	1.10	1.87	1.58	2.75	1.23	1.66
10	env($r=1$) NH0d3@N (c) (C) C	1.17	1.05	1.01	0.46	1.96	2.16
11	env($r=1$) CH1d3@C (C) (C) N	1.14	1.09	1.10	0.68	2.01	1.89
12	env($r=2$) OH0d2@O (CC) C (C) C	0.90	1.10	0.99	0.93	1.10	2.22

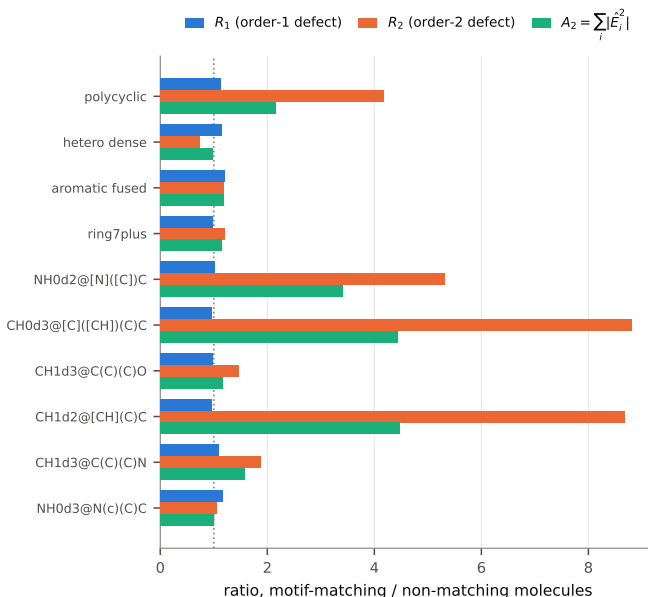


Figure 4: Order-resolved signature of each library motif. A motif that raises R_2 and A_2 but not R_1 is one whose energy the angular order cannot resolve.

detects a decomposition that has been asked an impossible question. It does not detect one that has confidently answered the wrong one. We take this to be the structural ceiling on any uncertainty estimate read off the hierarchy, and Section 6 shows it is where the confidently-wrong predictions live.

The premise behind T_{geo} holds up independently. The truncated energies $\hat{E}_{\text{trunc}}^k$ have MAEs of 20.454 ± 0.003 , 2.755 ± 0.052 and 0.778 ± 0.014 kcal/mol at $k = 0, 1, 2$ — a factor 7.4 from the dressed-atom model to first order and 3.5 from first to second. The decay is geometric with a slowing ratio rather than exactly geometric, which is why a tail estimator built on it helps (R plus T_{geo} beats R alone) without closing the gap on its own.

Table 6: Transfer of the motif library to ANI-1x. MAE_{in} and MAE_{out} are the mean per-class conformational MAE (kcal/mol) for connectivity classes that do and do not match the motif; “transfers” requires ratio > 1 and a Benjamini–Hochberg adjusted $q < 0.05$ on a one-sided Mann–Whitney test over classes.

#	Motif	classes	MAE_{in}	MAE_{out}	ratio	transfers
1	≥ 3 rings	674	26.643	20.122	1.32	yes
2	heteroatom fraction ≥ 0.4	7,958	19.361	21.765	0.89	no
3	fused aromatic	626	18.178	20.532	0.89	no
4	≥ 7 -membered ring	456	22.409	20.364	1.10	yes
7	$\text{env}(r=1)$ CH1d3@C(C)(C)O	33	20.040	20.430	0.98	no
9	$\text{env}(r=1)$ CH1d3@C(C)(C)N	33	24.426	20.420	1.20	yes
10	$\text{env}(r=1)$ NH0d3@N(c)(C)C	24	20.291	20.429	0.99	no
11	$\text{env}(r=1)$ CH1d3@C(C)(C)N	54	22.383	20.422	1.10	yes
13	$\text{env}(r=1)$ NH1d2@N(C)c	48	17.526	20.439	0.86	no
14	$\text{env}(r=1)$ CH1d2@C(=N)N	166	20.946	20.423	1.03	no
15	$\text{env}(r=1)$ CH1d3@C(C)(C)O	56	27.501	20.401	1.35	yes
18	$\text{env}(r=1)$ CH1d3@C(C)(C)O	20	27.692	20.419	1.36	yes

5.6 Transfer to non-equilibrium geometries

The first thing the ANI-1x evaluation says is not about motifs. A HIP-NN trained on QM9’s minimum-energy geometries has essentially no usable conformational energy surface. Over 2,120,509 conformers in 14,320 connectivity classes the mean error on within-class relative energies is 19.540 ± 0.422 kcal/mol against a median within-class spread of 27.75 kcal/mol, a ratio of 0.73; restricted to the thermally accessible stratum — conformers within 5.0 kcal/mol of their class mean, about what a 300–500 K trajectory explores — it is 9.545 ± 0.111 kcal/mol, still larger than the spread being predicted. Part of the reason is visible in the architecture rather than the chemistry: 30.38% of ANI-1x conformers contain an atom pair closer than the inner edge of the innermost sensitivity function (≈ 1.87 Bohr, from $R_{\text{low}} = 1.7$) and 1.92% closer than R_{low} itself, where the inverse-distance basis has no support at all. The comparison has a floor of its own, measured directly from the two levels of theory ANI-1x supplies: relative conformer energies at $\omega\text{B97X}/6\text{-}31\text{G(d)}$ and $\omega\text{B97X}/\text{def2-TZVPP}$ differ by 0.855 kcal/mol at the median class, 3.2% of the spread.

Against that background the library transfers partially, in a pattern that follows the mechanism exactly. Of the 18 library motifs testable on ANI-1x, 39% separate harder from easier chemistry over the full conformer range, and 67% of the 12 testable in the near-equilibrium stratum do. The motifs that transfer at every geometry are the skeletal ones: polycyclic frameworks keep a conformational-error ratio of 1.32 over the full range and 1.34 near equilibrium, and large rings likewise. The motifs that transfer only near equilibrium are the electronic ones — heteroatom loading and fused aromatics both sit below unity over the full conformer range and above it in the thermally accessible stratum. That is the expected signature: a polycyclic skeleton is polycyclic at every geometry, whereas a motif whose failure signature is an electronic effect at the equilibrium structure is swamped once the geometry itself becomes the dominant source of error.

The hierarchy signal transfers, weakly. Refitting only the isotonic stage on an ANI-1x discovery half and keeping the QM9 ranking coefficients, HiRE attains a rank correlation of 0.235 ± 0.004 with the conformational error at an expected calibration error of 0.063 ± 0.003 kcal/mol — a usable ordering on chemistry the model was never trained for, on a task where the model itself is close to uninformative.

6 Where the analysis fails

Two things in this study do not work, and both are informative.

What the motif library misses. $20.4 \pm 0.5\%$ of the > 1 kcal/mol failures on the validation half are not matched by any motif in the library. Those molecules are not a random remainder: their mean error is 1.49 ± 0.01 kcal/mol against 2.91 ± 0.04 kcal/mol for the failures the library does catch, and their mean non-hierarchicality is 0.114 ± 0.003 against 0.316 ± 0.014 . The library therefore captures

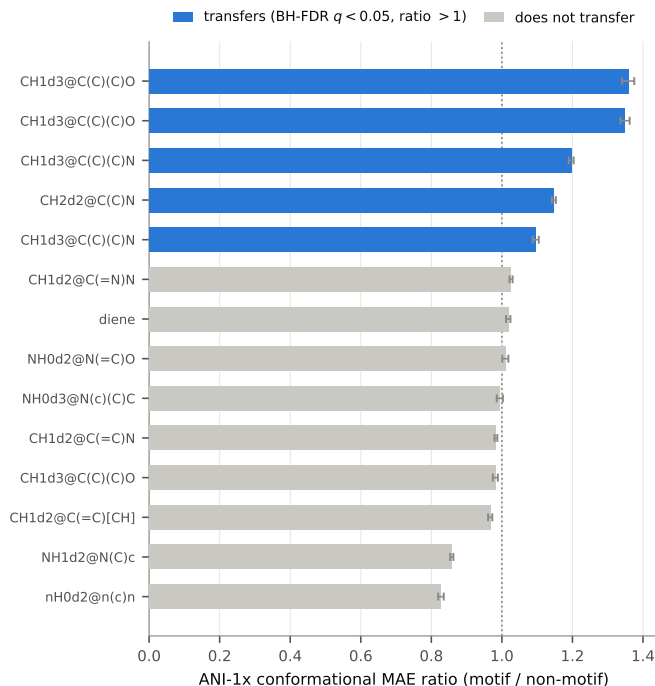


Figure 5: Per-motif conformational-error ratio on ANI-1x. Motifs to the right of unity flag molecules whose off-equilibrium energy surface HIP-NN also gets wrong.

the *large* failures and leaves behind a diffuse population that is barely over the threshold and whose hierarchy looks healthy. That is consistent with the compactness result rather than in tension with it: the error tail has a structured head and an unstructured shoulder, and 893 candidate predicates could not find structure in the shoulder. Whether the shoulder is genuinely structureless or merely invisible to a molecule-level substructure vocabulary is open; an atom-level analysis is the obvious next attempt, and QM9 offers no per-atom reference energy against which to validate one.

Where the uncertainty model fails, and why it is the same story. $20.8 \pm 0.9\%$ of all failures fall in the *lower half* of the HiRE ranking — the model is confidently wrong about them, with a mean error of 1.57 ± 0.02 kcal/mol. In the other direction, $25.3 \pm 0.4\%$ of the most-flagged decile have errors below 0.5 kcal/mol. Neither is noise. The motifs most enriched among the confidently-wrong molecules are the ones whose R_2 ratio sits near unity — large rings, and saturated carbon environments — at enrichments up to $2.7\times$. The motifs most enriched among the false alarms are the opposite kind: aromatic six-ring nitrogen ($4.2\times$) and the radical-valence carbon environments ($3.7\times$), where the hierarchy fires on every molecule in the class while only a fraction of them are actually wrong.

The two failure modes have a single cause: R and everything built on it measure whether the decomposition *struggles*, not whether it is *wrong*. A delocalised aromatic system that the model resolves badly but smoothly produces a decaying hierarchy and a large error. A polycyclic cage produces a non-decaying hierarchy whether or not the answer happens to come out right. Any uncertainty estimate read off the hierarchy inherits both. It is also why the ensemble remains better (Table 3): disagreement between independently trained models is sensitive to the second failure mode in a way that a single model’s internal decomposition structurally cannot be. Adding a signal that measures distance to the training distribution rather than internal strain — latent-space distance [Janet et al., 2019] is the obvious candidate — should attack exactly the hierarchy-blind population, and we would expect it to be complementary rather than redundant.

7 Cost

Training one HIP-NN on the 19,799-molecule extrapolation partition takes 5.6 s per epoch and 1.8 h in total on one NVIDIA H200, at the published minibatch size of 30; the published-protocol runs take 13.6 s and 30.2 s per epoch at the two larger training-set sizes. Evaluating the trained model on the 109,024 nine-heavy-atom molecules, including capture of every per-atom, per-order contribution, costs 0.062 ms per molecule.

The marginal cost of HiRE on top of that is essentially zero. The quantities in Table 1 are formed from tensors the forward pass has already produced — fourteen reductions over an $(N_{\text{atom}} \times 3)$ array — and Stage A is a dot product with fourteen coefficients. Fitting the model once on the discovery half takes under a second. This is the point of the design: the alternative that outperforms it, an 8-seed ensemble, costs 8 trainings (8×1.8 h here) and 8 forward passes for every molecule scored, forever. If a user is willing to pay that, the two combine; if not, HiRE recovers 25% of the gap between the paper’s signal and that ensemble for nothing.

Motif evaluation costs one RDKit parse and one substructure-match pass per molecule, 1.04 ms, which is dominated by the SMILES parse rather than by the matching. The geometry-to-graph correspondence is the only step with a failure mode, and it is rare (99.53% of QM9 succeeds).

8 Limitations

The fidelity runs sit a few per cent above the published MAE. Both published-protocol sizes terminate on the paper’s own stopping rule, but both land above the published numbers: 4% at $N_{\text{train}} = 49,000$ and 9% at 109,426. The remaining protocol difference is that we averaged 4 and 2 seeds where the paper averaged eight, so our means carry more sampling noise; beyond that we can identify no specific cause, and we prefer to leave it as an unexplained residual rather than assert one.

What that residual can and cannot contaminate is worth being precise about. The models used for *every* motif, calibration, compactness and transfer number in this paper are the 8 extrapolation-partition runs, which is the paper’s own N_{model} , and each of them ran a mean of 1167 epochs to the same stopping rule. Their in-distribution error, 0.338 ± 0.012 kcal/mol from 19,799 training molecules, is $0.64 \times$ the published learning curve extrapolated to that training size. A few per cent of unexplained error on two models we never analyse cannot manufacture the motif structure we report from 8 models we did.

The residual question is whether a *more accurate* model would show the same failure anatomy. We cannot settle it within the held-fixed constraint, and we do not claim to. The direction of the bias is the wrong one for our conclusions: a weaker model fails more diffusely across all chemistry, which would dilute motif enrichment and depress compactness, whereas we report concentrated, chemically coherent motifs and a compactness result that is already negative. A stronger model would plausibly show a lower absolute failure rate — which moves the compactness statistic, since attribution is defined against that rate — and we would expect the hierarchy-blind class of Section 5.5 to persist, because its cause is the two-bond receptive field rather than the amount of training.

Two deviations from the published loss. The L_2 term of Eq. (16) includes the energy-regression weights $\|w\|^2/\sigma_E^2$, which hippynn’s regularisation-parameter collection does not expose. Measured on the trained models, the omitted term would contribute 9.4×10^{-7} to the loss against 1.3×10^{-3} for the network term that *is* included, and against a total loss of order 10^{-2} — a ratio of 7.0×10^{-4} . We also suppress hippynn’s per-minibatch warning that a handful of QM9 O–H bonds fall just inside the innermost sensitivity function; that is a logging change, not a modelling one. Both are recorded in `codebase_notes.md` and `logs/12_term_check.json`.

The motif library is conditioned on one architecture and one split. Every motif here is a statement about a two-interaction-layer HIP-NN with 80 features extrapolating from eight to nine heavy atoms. The radius-2 environment vocabulary is tied to $N_{\text{int}} = 2$ by design, so a deeper model would need a different vocabulary; whether the same chemistry would fail a four-layer model is an open question that our held-fixed constraint deliberately prevents us from answering. Nor is the eight-to-nine split the only extrapolation that matters — element extrapolation, charge states and open-shell systems are all outside QM9.

The uncertainty model is post-hoc and marginal. HiRE estimates $\mathbb{E}[|\hat{E} - E|]$, not a predictive distribution, and its calibration guarantee is empirical rather than finite-sample. Conformal prediction [Hu et al., 2022] would supply coverage guarantees on top of the same score and is the obvious next layer. We also did not use latent-space distance to the training set, which is known to be a strong signal [Janet et al., 2019] and would likely add to HiRE; we left it out because it is a published method rather than a property of the hierarchy, and the question here is what the hierarchy can say about itself.

The conformational transfer test has a hard floor and a selection step. Relative conformer energies computed at ω B97X/6-31G(d) and ω B97X/def2-TZVPP differ by 0.855 kcal/mol at the median class, or 3.2% of the conformational spread, which bounds how sharply any transfer statement can be made. And 50.6% of the scanned conformers survive the connectivity-class step; the discarded ones are those whose perceived bonding differs from their group’s majority, i.e. geometries near bond breaking, so the test is run on the better-behaved part of ANI-1x sampling.

9 Conclusions

The hierarchical decomposition that HIP-NN was built around does contain a usable self-diagnosis, and it is worth more than the original scalar extracts from it. Separating the pairwise defect from the angular one, restoring the kcal/mol scale the ratio discards, and extrapolating the geometric tail of the many-body expansion raises the AUROC for catching > 1 kcal/mol failures from 0.747 ± 0.006 to 0.779 ± 0.004 on chemistry a heavy-atom larger than anything the model was trained on, at no cost beyond a single forward pass it was already making.

What the decomposition diagnoses, though, is narrower than “error”. The failure motifs whose error the hierarchy sees are those where the local environment does not determine the local energy — polycyclic skeletons, and molecules whose bonding is ambiguous enough that automated perception assigns a radical valence, where the atoms carrying the ambiguity carry $7.7\times$ the order-2 defect of their neighbours in the same molecule. The failure motifs it does not see are the heteroatom-electronic ones: molecules loaded with N, O and F, adjacent-heteroatom bonds and heteroaromatic rings, whose energetics depend on lone-pair and π interactions reaching further than two bonds and which the model gets wrong smoothly. 24% of the screened motifs are of this second kind, and they are where the confidently-wrong predictions concentrate. That is a structural statement about non-hierarchicality as an uncertainty signal, not a defect of our particular estimator: a measure of whether a decomposition strains cannot detect a decomposition that is comfortably wrong.

Two further conclusions are worth separating from the headline because they are negative. The failure modes of this model are concentrated but not compact — five nameable chemical classes carry 76.9% of the failures at $1.69\times$ enrichment, and no small library we could construct from 893 candidates reaches 80% without flagging half the test set. And a motif library identified on minimised geometries transfers only partly off equilibrium (67% of the testable library motifs near equilibrium, 39% over the full ANI-1x conformer range), with the skeletal motifs transferring at every geometry and the electronic ones only near equilibrium — a distinction that matters, though it is dwarfed by the observation that the underlying model has almost no conformational energy surface to triage in the first place.

The practical recommendation that follows is narrow and, we think, defensible. For screening minimised structures with a HIP-NN-style model, compute HiRE — it is free, it is calibrated in kcal/mol, and flagging the least reliable 10.2% of predictions recovers $30.9 \pm 0.5\%$ of the failures. Do not expect it to catch the aromatic ones, and do not expect a motif library built at equilibrium to survive a trajectory.

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A Reimplementation detail

Table 7: All HIP-NN runs. “valid. MAE” is the best validation MAE reached (the quantity early stopping tracks); “test MAE” is the held-out split of the corresponding protocol — for the extrapolation runs that is the in-distribution ≤ 8 -heavy-atom split, not the nine-heavy-atom set. Per-epoch and per-run records are in logs/runs/.

Run	seeds	N_{train}	epochs	s/epoch	valid. MAE	test MAE
≤ 8 heavy atoms	8	19,799	1167	5.6	0.334 ± 0.012	0.338 ± 0.012
≤ 8 heavy, hippynn default init	1	19,799	1314	6.4	0.338	0.333
≤ 8 heavy, default init, 300-epoch pilot	1	19,799	300	6.8	0.534	0.573
random QM9, $N_{\text{tr+val}} = 50,000$	4	49,000	1106	13.6	0.364 ± 0.010	0.367 ± 0.009
random QM9, $N_{\text{tr+val}} = 110,426$	2	109,426	1207	30.2	0.262 ± 0.009	0.279 ± 0.002

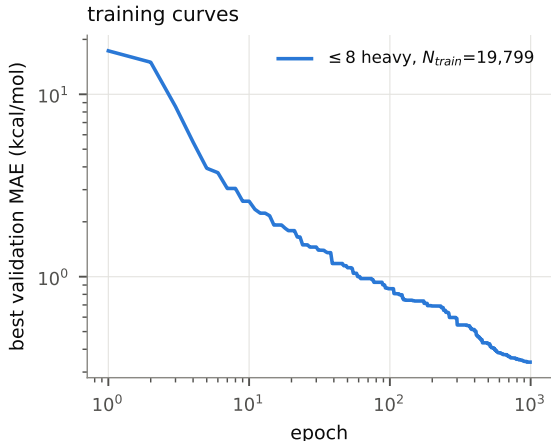


Figure 6: Best-so-far validation MAE against epoch, averaged over seeds within each protocol.

B Reproduction of the original uncertainty observation

C Complete screened-motif table

Table 4 in the main text lists the greedy library. The complete set of 233 screened motifs with their validation statistics, including the 20 that do not survive false-discovery-rate correction, is distributed as logs/analysis/analysis.json (field motif_table) and summarised for the named-chemistry family below.

D What the environment motifs denote

Morgan environment identifiers are hashes, so Table 4 names them by the fragment SMILES of the environment and the central atom’s label (element $n_H n_{\text{heavy}}$). Table 9 gives, for each library motif, one matching molecule drawn at random from the *discovery* half, so that the reader can see the chemistry the identifier picks out without inspecting held-out data.

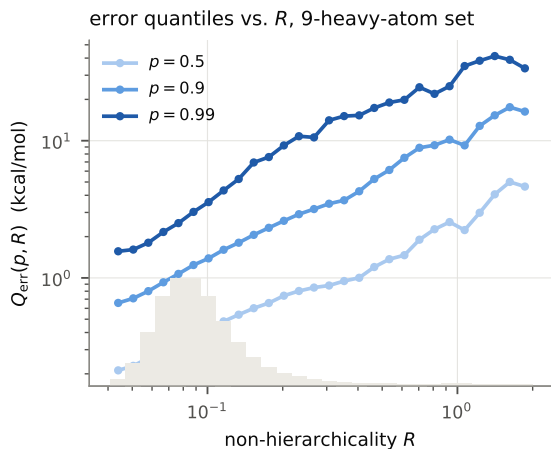


Figure 7: Error quantiles conditioned on the non-hierarchy R for our reimplement on the nine-heavy-atom set, in the format of Fig. 3 of Lubbers et al. [2018]. The grey histogram is the distribution of $\log_{10} R$. The monotone rise of every quantile with R reproduces the original observation on chemistry the original paper did not test.

Table 8: All named-chemistry predicates that passed the discovery screen, with their validation-half statistics, whether or not they reached significance. Environment motifs are omitted here for space and are in the JSON record.

Motif	cov. (%)	$\langle \Delta E \rangle$	lift	AUC	attr. (%)	q_{BH}
≥ 3 rings	24.8	1.45	2.61	0.648	44.1	$< 10^{-12}$
unseen local chemistry (class)	17.7	1.44	2.27	0.624	30.8	$< 10^{-12}$
heteroatom fraction ≥ 0.4	17.4	0.87	1.15	0.590	25.6	$< 10^{-12}$
$\geq$ half of atoms in unseen radius-2 environments	10.6	1.90	2.96	0.657	22.1	$< 10^{-12}$
π -bond density ≥ 0.5	15.2	0.83	1.09	0.558	20.6	$< 10^{-12}$
unseen radius-1 environment	12.3	1.22	1.70	0.608	20.4	$< 10^{-12}$
fused or N/O heteroaromatic (class)	14.6	0.83	1.08	0.556	19.7	$< 10^{-12}$
ambiguous perceived bonding	8.5	1.65	2.37	0.631	16.1	$< 10^{-12}$
ambiguous perceived bonding (class)	8.5	1.65	2.37	0.631	16.1	$< 10^{-12}$
aromatic 5-ring	11.9	0.84	1.10	0.551	16.0	$< 10^{-12}$
adjacent heteroatoms	8.9	0.96	1.26	0.585	13.9	$< 10^{-12}$
adjacent heteroatoms (class)	8.9	0.96	1.26	0.585	13.9	$< 10^{-12}$
aromatic 5-ring, 2 heteroatoms	6.1	0.85	1.10	0.561	8.7	$< 10^{-12}$
fused aromatic	2.8	1.62	2.16	0.742	8.2	$< 10^{-12}$
adjacent aromatic N-N	4.9	0.93	1.21	0.583	7.8	$< 10^{-12}$
bond stretched $> 1.15 \times$ covalent	3.9	1.82	2.48	0.624	7.5	$< 10^{-12}$
≥ 7 -membered ring	3.2	1.50	2.00	0.672	7.1	$< 10^{-12}$
≥ 7 -membered ring (class)	3.2	1.50	2.00	0.672	7.1	$< 10^{-12}$
aromatic 6-ring N	3.0	1.06	1.39	0.592	4.8	$< 10^{-12}$
geminal heteroatom substitution (class)	3.0	1.04	1.36	0.577	4.5	$< 10^{-12}$
adjacent aromatic N-O	2.6	0.97	1.26	0.590	4.2	$< 10^{-12}$
acetal	2.6	1.03	1.34	0.579	4.0	$< 10^{-12}$
N-N-N chain	1.6	1.14	1.48	0.616	3.0	$< 10^{-12}$
N-O single bond	1.9	0.99	1.29	0.572	2.8	1.5×10^{-9}
enamine	1.5	1.05	1.35	0.601	2.4	$< 10^{-12}$
N-N single bond	0.3	2.09	2.70	0.735	0.8	$< 10^{-12}$
1,3-diene	0.4	1.34	1.73	0.651	0.7	5.2×10^{-9}

E Compactness under alternative orderings

Table 9: One example molecule for each library motif, as the canonical SMILES of the structure QM9 perceives from the relaxed geometry. Radical-like valences ([CH], [N], ...) in these strings are the perception failures that the *ambiguous perceived bonding* predicate detects.

#	Motif	Example matching molecules (discovery half)
1	≥ 3 rings	<chem>CC(=O)N[C@H]1[C@H]2C[C@@H]1C2</chem>
2	heteroatom fraction ≥ 0.4	<chem>O=CC(O)(C=O)[C@H]1CN1</chem>
3	fused aromatic	<chem>O=C1Cc2[nH]nnc2N1</chem>
4	≥ 7 -membered ring	<chem>C[C@H]1[C@@H]2CCOC[C@H]1C2</chem>
5	$\text{env}(r=1)$ <chem>NH0d2@[N]([C])C</chem>	<chem>C[C@H]1O[C]2[N]CC(=O)[C@H]21</chem>
6	$\text{env}(r=1)$ <chem>CH0d3@[C]([CH])(C)C</chem>	<chem>[CH]1C[C@@H]2[C@@H]3C=CC[C]1[C@@H]32</chem>
7	$\text{env}(r=1)$ <chem>CH1d3@C(C)(C)O</chem>	<chem>O=C1[C@H]2CC[C@]1(O)CO2</chem>
8	$\text{env}(r=1)$ <chem>CH1d2@[CH](C)C</chem>	<chem>O[C@@H]1[C@H]2[C]3[CH]CO[C@H]3[C@@H]12</chem>
9	$\text{env}(r=1)$ <chem>CH1d3@C(C)(C)N</chem>	<chem>C1[C@@H]2O[C]3[N][C@H]1[C@@H]2N31</chem>
10	$\text{env}(r=1)$ <chem>NH0d3@N(c)(C)C</chem>	<chem>Cc1cn[nH]c1N1CC1</chem>
11	$\text{env}(r=1)$ <chem>CH1d3@C(C)(C)N</chem>	<chem>Cn1nnc([C@H]2CN2)n1</chem>
12	$\text{env}(r=2)$ <chem>OH0d2@O(CC)C(C)C</chem>	<chem>O=C1CCO[C@H]2CC[C@@H]12</chem>

Table 10: Compactness under four orderings of the same screened motif set: the lift-corrected greedy objective used in the main text; the uncorrected objective that maximises raw failure coverage; and the lift-corrected objective restricted to the chemistry-class groupings and to single predicates respectively. The uncorrected objective is what a literal reading of “the top K motifs cover 80% of failures” asks for, and the “molecules matched” column is why we do not report it as the headline.

Ordering	K_{80}	attr. at $K=5$ (%)	molecules matched at $K=5$ (%)	enrichment at $K=5$
lift-corrected greedy (main text)	not reached	76.9	45.5	1.69
raw failure coverage	5	83.6	56.4	1.48
chemistry-class groupings only	not reached	52.8	33.4	1.58
single predicates only	not reached	76.9	45.5	1.69